ELLab

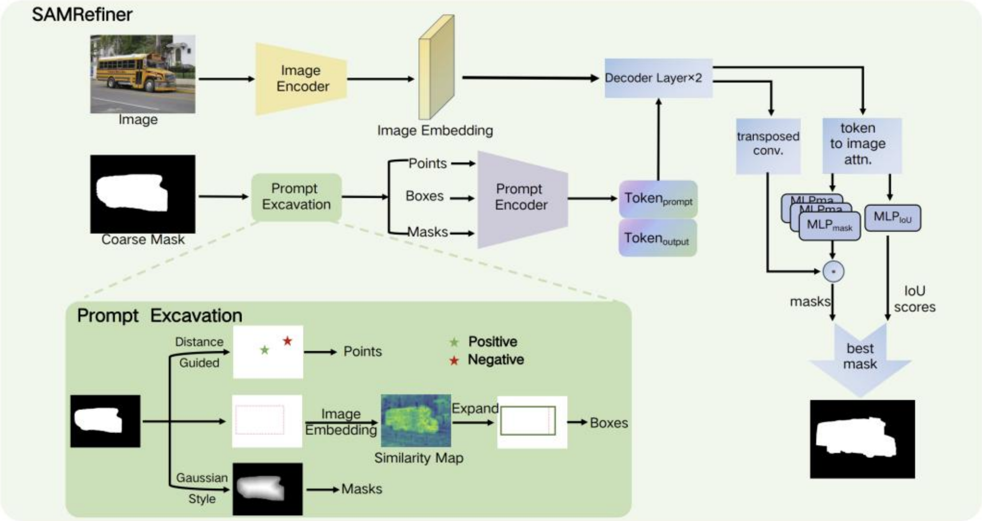
SUNGKYUNKWAN UNIVERSITY

Anomaly Segmentation (+Diffusion)

2026.05.26

SAMRefiner

Method



Multi-Prompt Excavation Framework

- SAM refinement 성능은 input prompt quality에 크게 의존함
- Coarse mask는 noise와 defect를 포함, naive prompt 생성은 불안정함

Core Idea

- coarse mask로부터 point / box / mask prompt를 생성하고 상호 보완적으로 활용하여 refinement 성능 향상

Point Prompt

- Positive point: foreground 내부에서 background와의 distance가 최대인 point 선택 (distance transform 기반)
- Negative point: foreground와 가장 멀리 떨어진 background point 선택 & foreground bbox 내부에서 선택하여 localization 강화

SAMRefiner

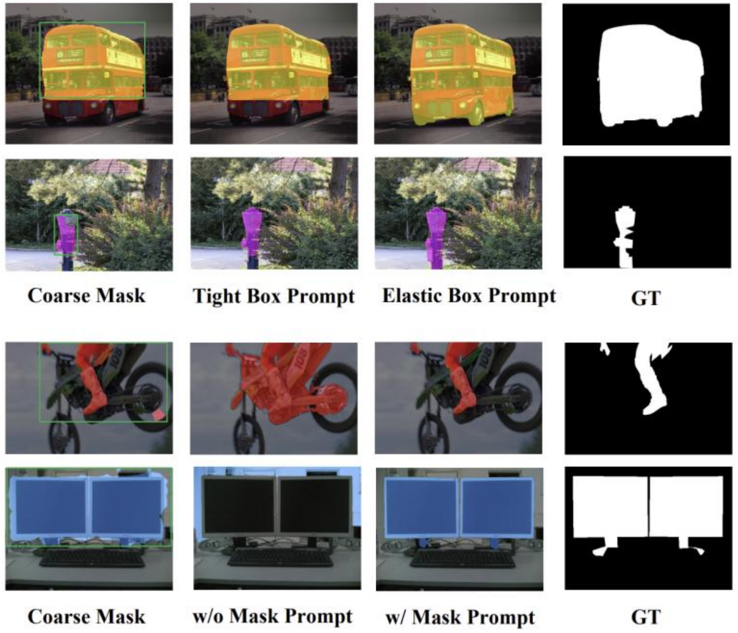
Method

Box Prompt (CEBox)

- coarse mask 기반 tight box

→ false negative 영역을 포함하지 못할 수 있음

- SAM image embedding feature 기반 similarity 계산
- foreground feature와 유사한 방향으로 left/right/up/down 점진적 확장
- context-aware elastic box 생성



Mask Prompt

- mask prompt는 원래 iterative refinement용 auxiliary input으로 학습됨 → binary mask prompt alone에서 성능이 낮음
- coarse mask를 직접 사용하지 않고 distance-transform 기반 Gaussian mask 생성
- center region은 high confidence, boundary region은 low confidence로 표현

SAMRefiner

Method

Split-Then-Merge (STM)

- semantic segmentation mask는 category-wise mask이므로,
  하나의 mask 안에 여러 object가 포함될 수 있음
- SAM은 넓게 분포된 multiple objects segmentation에 어려움을 보임

Split

- coarse mask를 connected component 기준으로 분리
- noisy / trivial region 제거

Merge

- 가까운 region들을 iterative하게 병합
- merged box area 변화가 작고 merged box 내부 mask occupancy가 충분할 때

SAMRefiner

Method

IoU Adaptation

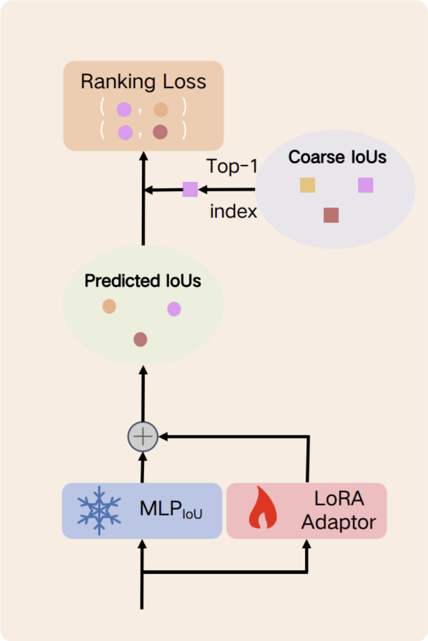
- SAM은 multiple mask candidates를 생성하지만, IoU prediction이 부정확함
- coarse mask를 prior로 활용하여 SAM IoU ranking 능력 개선
- $IoU_{coarse} = IoU(SAMoutputmask, coarsemask)$
- coarse IoU가 높은 mask를 intended mask로 간주

Training Strategy

- GT annotation 없이 self-boosted 방식으로 학습
- pretrained SAM은 freeze & IoU head에만 LoRA adaptor 추가

Ranking-based Loss

- regression 대신 ranking loss 사용
- coarse IoU가 가장 높은 mask가 가장 높은 IoU score를 갖도록 학습



(b) IoU Adaption

$$loss = \sum_{i=1, i \neq j}^{n} \max(0, x_i - x_j + m)$$

SAMRefiner

Application

Distance-guided point

- Gt 이용해서 upperbound 만들때 사용
- Negative point 추가 → foreground bbox 내부 background point

Multi-Prompt Collaboration

- mask alone는 약함 point + box + mask 조합이 strongest

STM (Split-Then-Merge)

- Broken/ crack 처럼 fragmented region 존재
- connected component 분리하고 가까운 fragment 병합하는 방식으로 응용

Experiment

Strong upper bound

SAMRefiner + ground truth

- GT → CEBox + positive/negative point extraction
- Point + Box + Mask prompt input to SAM
- Generate 3 candidate masks
- Oracle selection using maximum GT IoU
- Achieves strong upper-bound performance

	count	mean	std	min	max
bottle	63	0.828385	0.140696	0.403736	**0.992843**
cable	92	0.751488	0.207583	0.188940	**0.985802**
capsule	109	0.672356	0.179403	0.153809	**0.948826**
carpet	89	0.676817	0.171157	0.282096	**0.924826**
grid	57	0.552893	0.306435	0.034101	**0.936592**
hazelnut	70	0.830121	0.085448	0.510049	**0.965631**
leather	92	0.742227	0.156071	0.206868	**0.941987**
metal_nut	93	0.864249	0.105791	0.301424	**0.960320**
pill	141	0.798896	0.128128	0.203888	**0.987960**
screw	119	0.688820	0.237790	0.075025	**0.968559**
tile	84	0.878823	0.117286	0.084950	**0.985210**
toothbrush	30	0.613669	0.235818	0.106093	**0.951337**
transistor	40	0.632576	0.175137	0.310265	**0.957784**
wood	60	0.712131	0.133361	0.402898	**0.928042**
zipper	119	0.562318	0.234717	0.014190	**0.898990**

Experiment

Finetune upper bound

SAMRefiner + ground truth + LoRA Fine-tuning

- GT → CEBox + positive/negative point extraction
- Point + Box + Mask prompt input to SAM
- Fine-tune only the SAM mask decoder (LoRA adaptation)
- Supervise predicted mask using GT mask loss (BCE + Dice)
- Generate a single refined mask without Oracle selection

	count	mean	std	min	max
bottle	51	0.802571	0.177351	0.381794	**0.984409**
cable	74	0.725331	0.198312	0.171360	**0.980700**
capsule	87	0.663349	0.170939	0.124325	**0.953241**
carpet	72	0.732085	0.161065	0.306372	**0.931252**
grid	47	0.509429	0.320379	0.000000	**0.941477**
hazelnut	56	0.838126	0.092161	0.539471	**0.972355**
leather	73	0.740355	0.160200	0.301569	**0.947868**
metal_nut	74	0.847513	0.134871	0.287744	**0.968236**
pill	113	0.795879	0.139586	0.182933	**0.988889**
screw	94	0.678580	0.243631	0.048017	**0.961943**
tile	67	0.884849	0.092149	0.502367	**0.988322**
toothbrush	24	0.595182	0.220607	0.116030	**0.924755**
transistor	32	0.599389	0.176230	0.345253	**0.953360**
wood	48	0.722385	0.166777	0.138616	**0.950797**
zipper	96	0.562761	0.266432	0.035933	**0.918480**

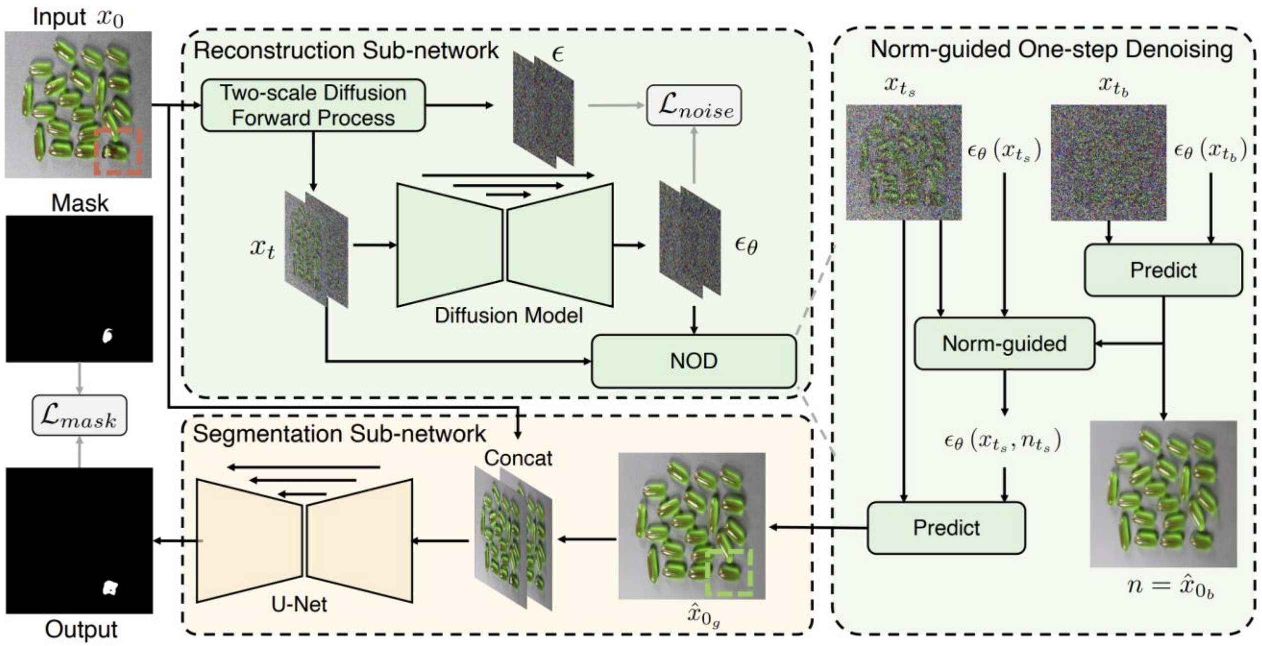
DiffusionAD

Reconstruction Sub-network (Noise-to-Norm)

- Noise-to-Norm paradigm using diffusion
- Remove anomaly information through Gaussian perturbation
- Recover anomaly-free image

Segmentation Module

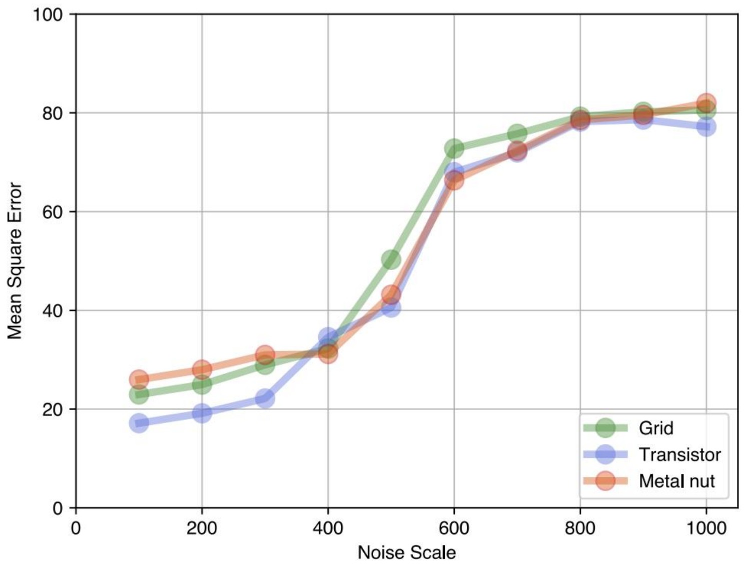
- Compare original and reconstructed images
- Learn inconsistencies and commonalities
- Predict pixel-level anomaly map

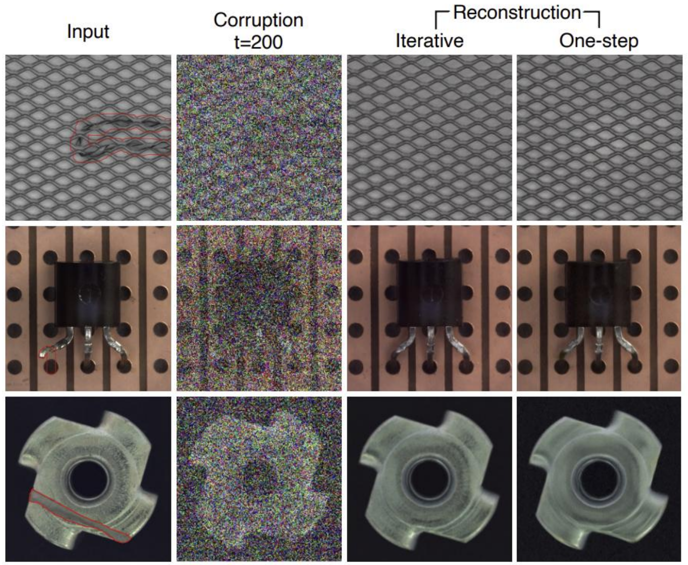


DiffusionAD

Observation I: Small noise scale enables efficient one-step reconstruction

- Iterative diffusion은 매 step마다 network inference 수행
- 작은 noise scale에서는 one-step reconstruction이 iterative 방식과 유사한 품질 유지
- t≤400 구간에서 reconstruction error(MSE)가 낮게 유지됨
- One-step denoising을 통해 계산량 및 inference 시간 크게 감소
- $\widehat{x_0} = \frac{1}{\sqrt{\overline{\alpha_t}}}\left(x_t - \sqrt{1-\overline{\alpha_t}}\epsilon_\theta(x_t, t)\right)$





DiffusionAD

Observation II: Different anomalies require different noise scales

Small anomaly (fine details)

- 작은 noise scale t_s사용
- Fine-grained details 보존
- 더 자연스러운 reconstruction 생성

Large anomaly / semantic anomaly

- 큰 noise scale t_b사용
- Anomalous regions를 더 강하게 perturb
- Semantic-level restoration 성능 향상

→ Norm-guide fusion

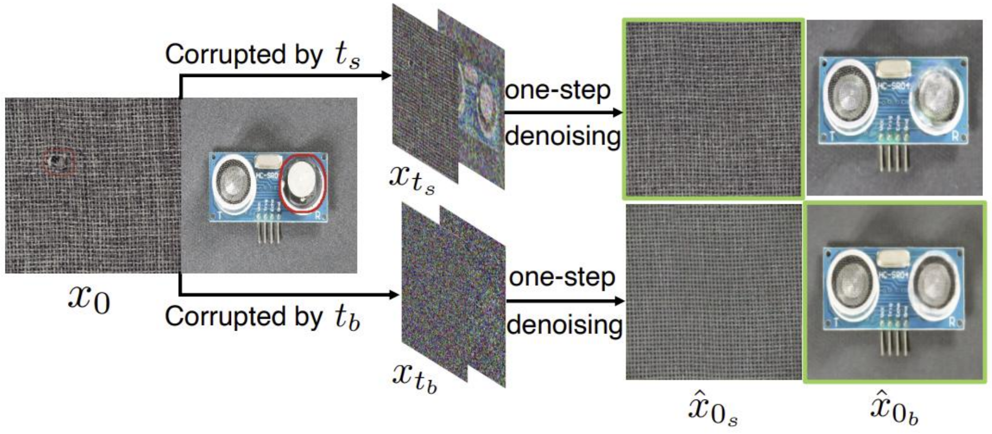
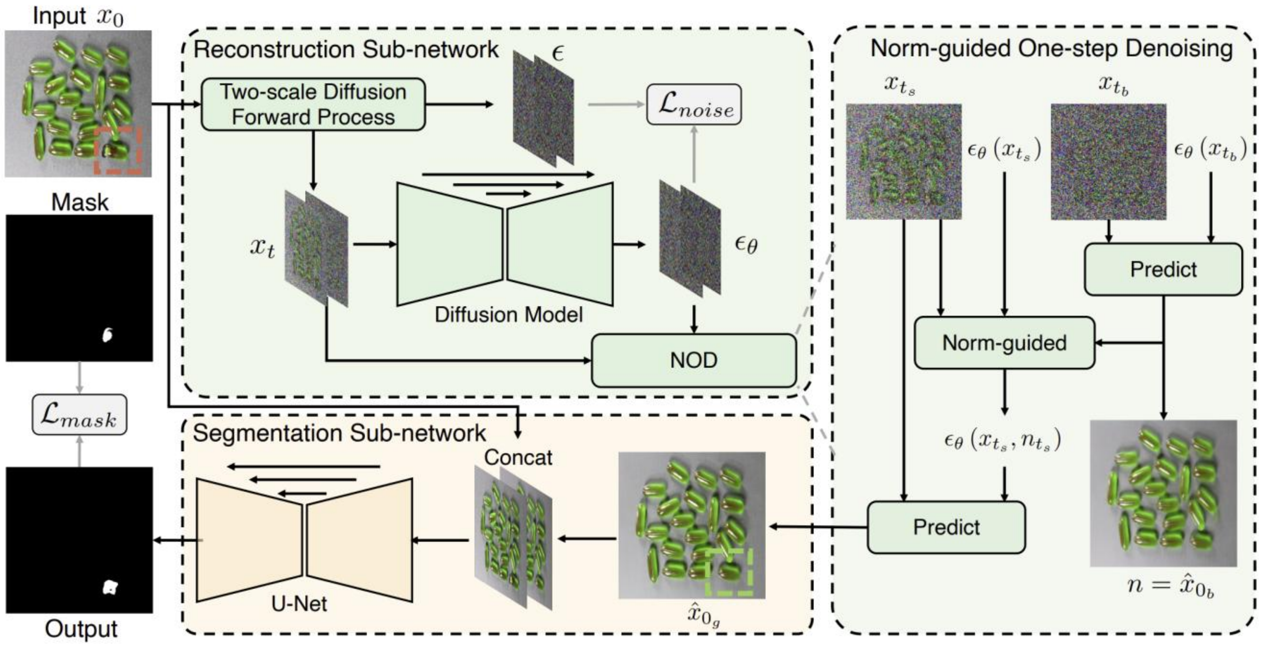


Fig. 5: **Observation II.** The impact of two different noise scales ($t_s = 200$ and $t_b = 400$) on the anomaly-free reconstruction for different types of anomalies. Images with green borders represent superior reconstructions.

DiffusionAD

Multi-scale reconstruction

- 입력 이미지 x_0에 두 개의 noise scale 적용
  - Small noise (t_s): $x_0 \rightarrow x_{t_s}$
  - Large noise (t_b): $x_0 \rightarrow x_{t_b}$
- Independent reconstruction
  - $x_{t_s} \rightarrow \hat{x}_{0s}$
  - $x_{t_b} \rightarrow \hat{x}_{0b} = n$
- Guidance generation
  - n을 다시 t_s로 perturb하여 n_{t_s}생성
  - Small-noise reconstruction을 anomaly-free 방향으로 유도
- Norm-guided one-step denoising
- 최종 anomaly-free reconstruction $\hat{x}_{0g}$생성



$$\epsilon_\theta(x_{t_s}, n_{t_s}) = \epsilon_\theta(x_{t_s}) - \sqrt{1-\bar{\alpha}_{t_s}}\; w\,(n_{t_s} - x_{t_s}) \qquad (7)$$

$$\hat{x}_{0_g} = \frac{1}{\sqrt{\bar{\alpha}_{t_s}}}\left(x_{t_s} - \sqrt{1-\bar{\alpha}_{t_s}}\,\epsilon_\theta\left(x_{t_s}, n_{t_s}\right)\right). \qquad (8)$$

DiffusionAD

Training Stage

Reconstruction Sub-network Training

- Normal pattern distribution 학습
  - Normal: y=0 / Anomalous: y=1

$$\mathcal{L}_{noise} = \frac{(1-y)(\|\epsilon_{t_s} - \epsilon_\theta(x_{t_s})\|^2 + \|\epsilon_{t_b} - \epsilon_\theta(x_{t_b})\|^2)}{2}. \quad (9)$$

$$\mathcal{L}_{mask} = \mathrm{Smooth}_{\mathcal{L}1}\left(M, \hat{M}\right) + \gamma \mathcal{L}_{focal}\left(M, \hat{M}\right). \quad (10)$$

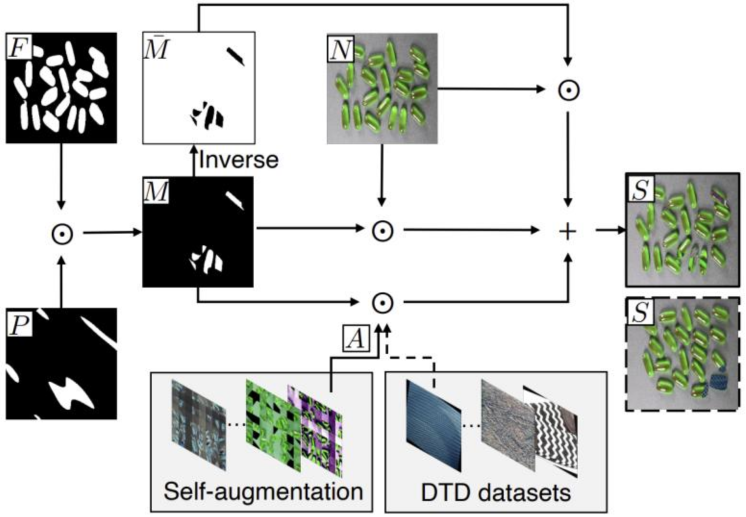
$$\mathcal{L}_{total} = \mathcal{L}_{noise} + \mathcal{L}_{mask}. \quad (11)$$

Segmentation Sub-network Training

- Input: Original image x_0, reconstruction image $\hat{x}_{0g}$
- Output: $\widehat{M}$ using Ground truth M

Synthetic Anomaly Generation

- Anomalous img의 gt mask 대신 normal img에 irregular anomaly region을 생성하고 gt mask를 생성함
- B는 anomaly와 normal을 섞는 비율

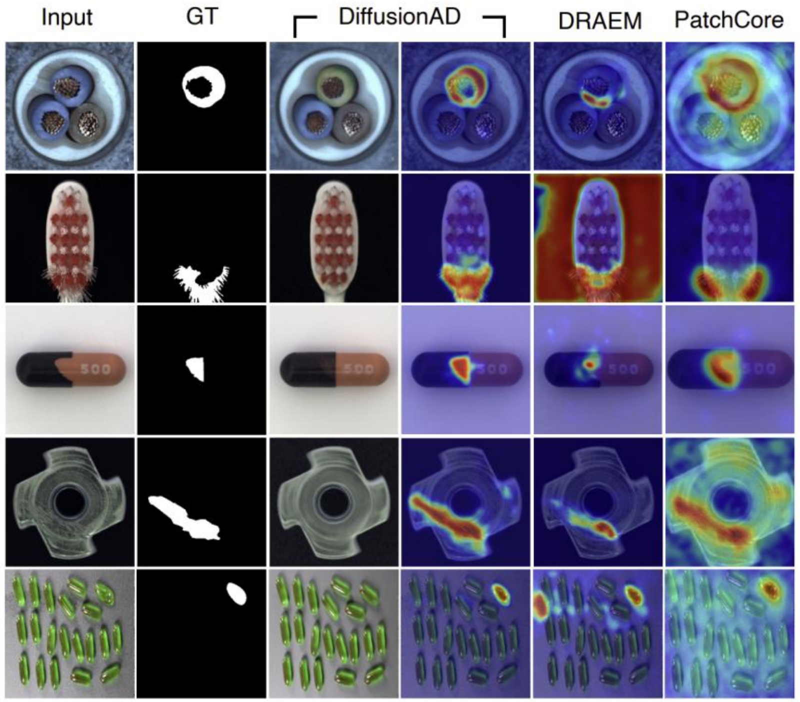


$$S = \beta(M \odot N) + (1-\beta)(M \odot A) + \bar{M} \odot N \quad (12)$$

DiffusionAD

Experiment

Category	Feature embedding-based				Generative model-based			
	PatchCore [2]	RD4AD [7]	RD++ [8]	SimpleNet [4]	DMAD [12]	DRAEM [3]	FastFlow [26]	**Ours**
Carpet	98.7/99.0	98.9/98.8	**100/99.2**	99.0/98.4	**100**/99.1	97.0/95.5	87.5/98.0	99.3/99.0
Grid	98.2/98.7	**100**/97.0	**100**/99.3	97.8/98.9	**100**/99.2	99.9/**99.7**	88.1/93.5	**100/99.7**
Leather	**100**/99.3	**100**/98.6	**100**/99.4	99.3/98.0	**100**/99.5	**100**/98.6	96.1/93.0	**100/99.8**
Tile	98.7/95.6	99.3/98.9	99.7/96.6	99.7/96.6	**100**/96.0	99.6/99.2	63.2/96.1	99.9/**99.7**
Wood	99.2/95.0	99.2/**99.3**	99.3/95.8	99.9/93.2	**100**/95.5	99.1/96.4	90.8/95.9	99.9/96.7
Bottle	**100**/98.6	**100**/99.0	**100**/98.8	**100**/94.0	**100**/98.9	99.2/**99.1**	**100**/92.7	99.7/**99.1**
Cable	99.5/98.4	95.0/**99.4**	99.2/98.4	**99.9**/98.7	99.1/98.1	91.8/94.7	91.2/97.5	99.6/98.1
Capsule	98.1/**98.8**	96.3/97.3	99.0/**98.8**	**100**/97.4	98.9/98.3	98.5/94.3	99.2/97.4	98.9/98.3
Hazelnut	**100**/98.7	99.9/98.2	**100**/99.2	**100**/98.6	**100**/99.1	**100/99.7**	98.8/98.9	**100/99.7**
Metal nut	**100**/98.4	**100**/99.6	**100**/98.1	**100**/97.0	**100**/97.7	98.7/99.5	90.8/98.1	**100/99.7**
Pill	96.6/97.4	99.6/95.7	98.4/98.3	**100**/98.8	97.3/98.7	98.9/97.6	98.0/92.3	99.0/**99.3**
Screw	98.1/99.4	97.0/99.1	98.9/**99.7**	**100**/98.0	**100**/99.6	93.9/97.6	97.8/99.4	99.4/99.0
Toothbrush	**100**/98.7	99.5/93.0	**100**/99.1	98.1/99.2	**100/99.4**	**100**/98.1	**100**/92.7	**100**/98.8
Transistor	**100**/96.3	96.7/95.4	98.5/94.3	**100/97.8**	98.7/95.4	93.1/90.9	88.1/94.3	99.8/92.9
Zipper	99.4/98.8	98.5/98.2	98.6/98.8	**100**/99.2	99.6/98.3	**100**/98.8	67.5/93.4	**100/99.2**
Average	99.1/98.1	98.5/97.8	99.4/98.3	99.6/97.6	99.6/98.2	98.0/97.3	90.5/95.5	**99.7/98.7**

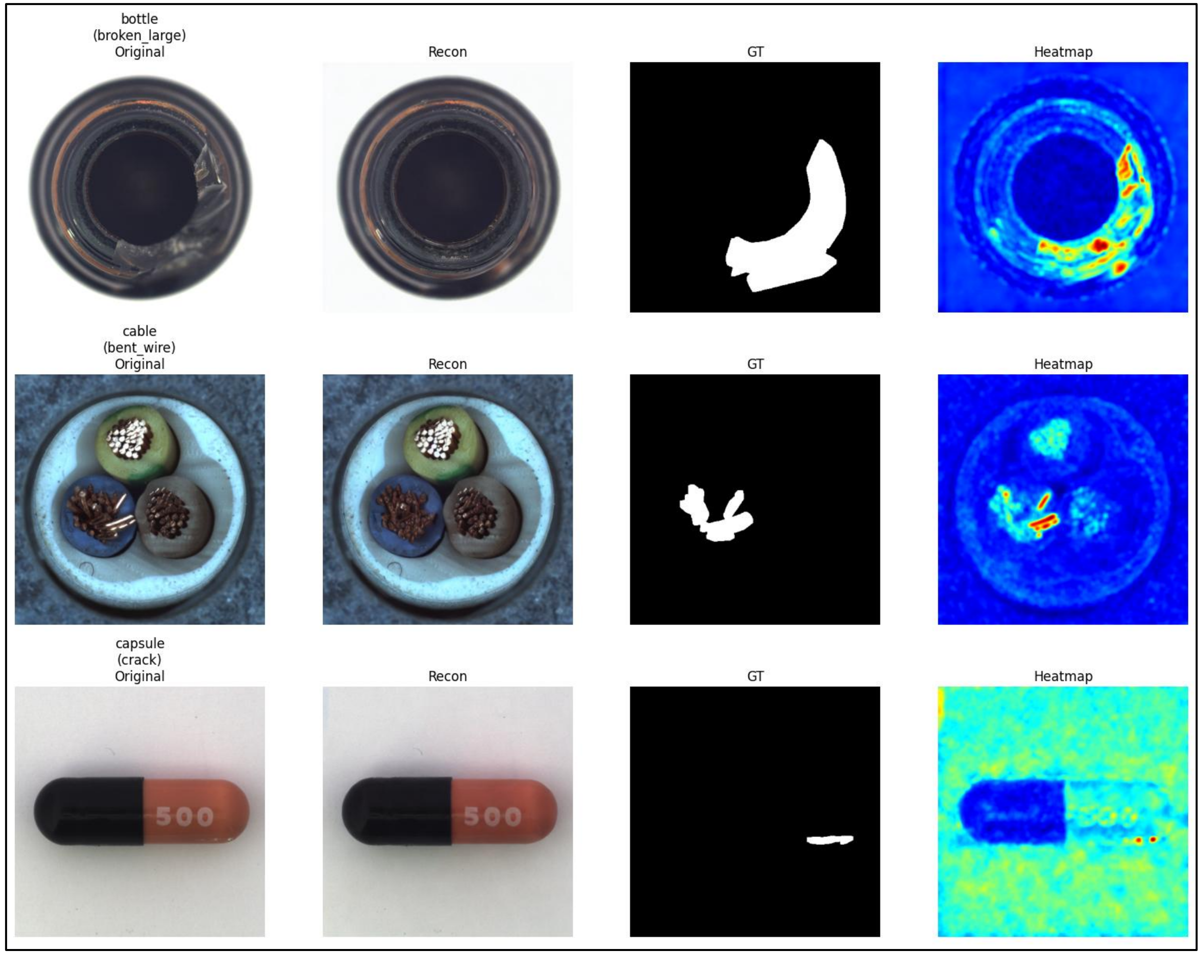


- DiffusionAD는 MVTec AD에서 가장 높은 평균 성능 (99.7 Image AUROC / 98.7 Pixel AUROC)을 달성함

Reconstruction

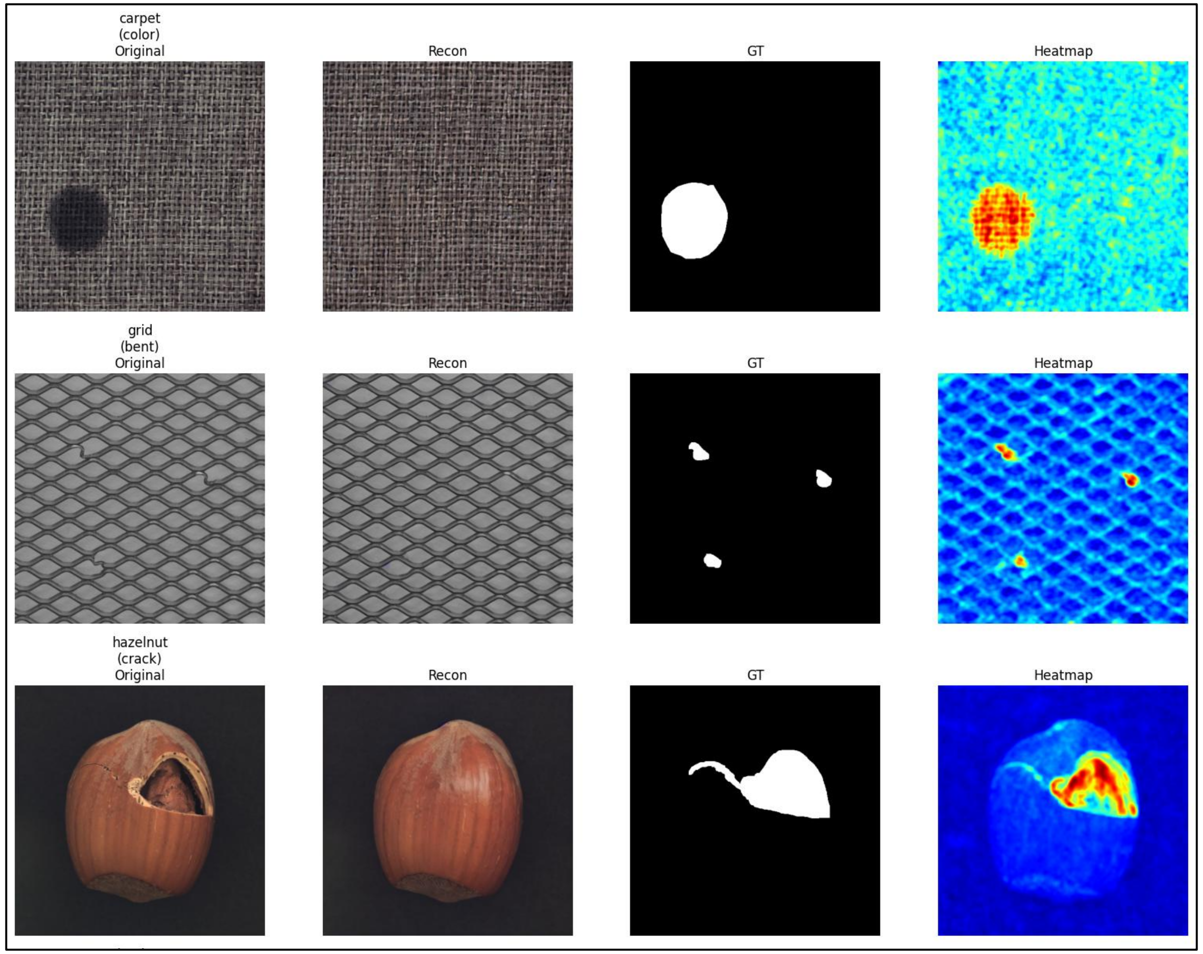
Method & results

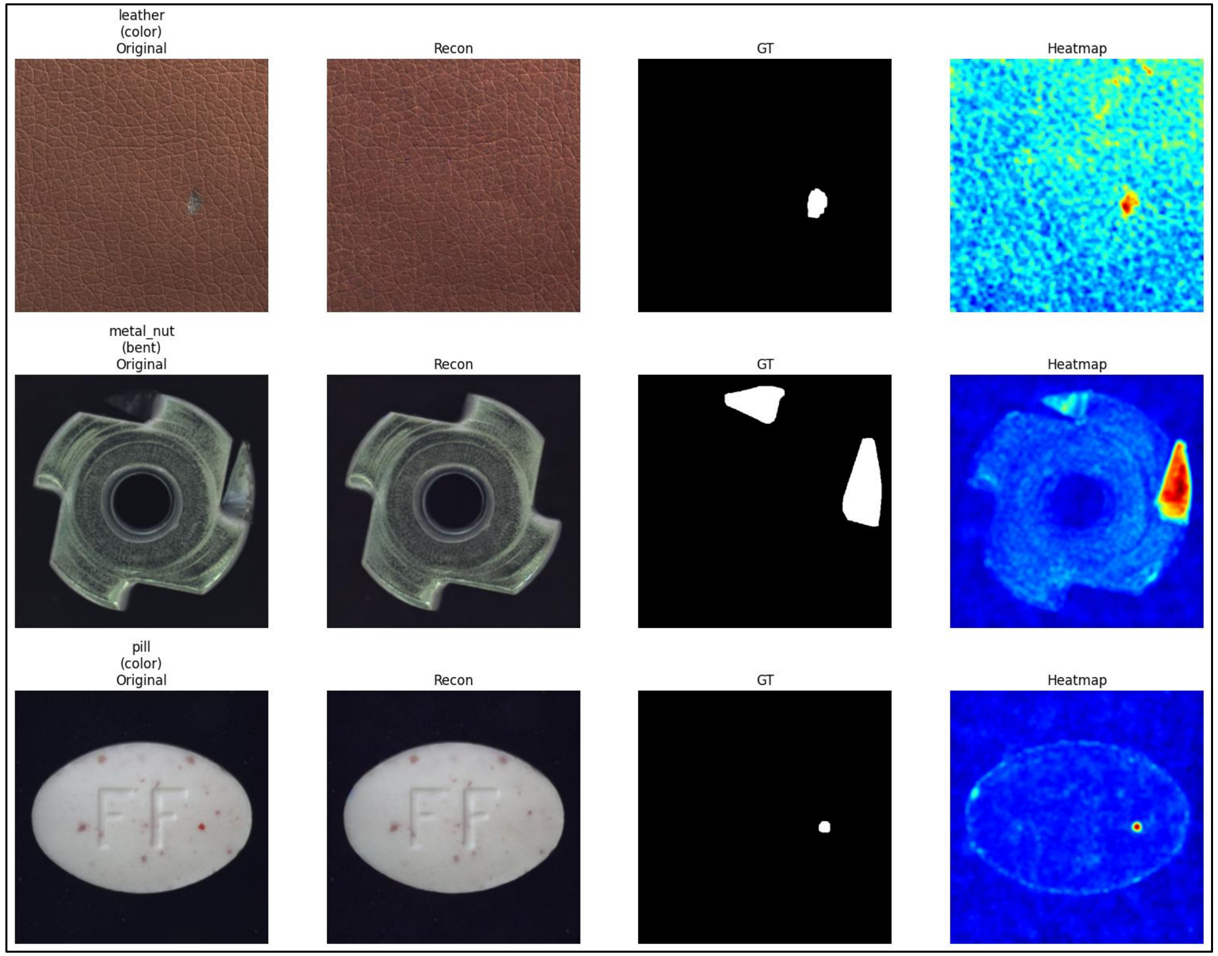
- 추가 학습 없이 pretrained Stable Diffusion Inpainting 모델 사용
- Oracle GT mask를 이용해 anomaly 영역 제거 (mask dilation 적용)
- 주변 정상 문맥 정보를 기반으로 masked 영역 Reconstruction 수행
- Original image와 Reconstruction 결과의 차이를 이용해 Difference Heatmap 생성
- Diffusion model이 anomaly 영역을 normal appearance로 복원 가능한지 확인



Reconstruction

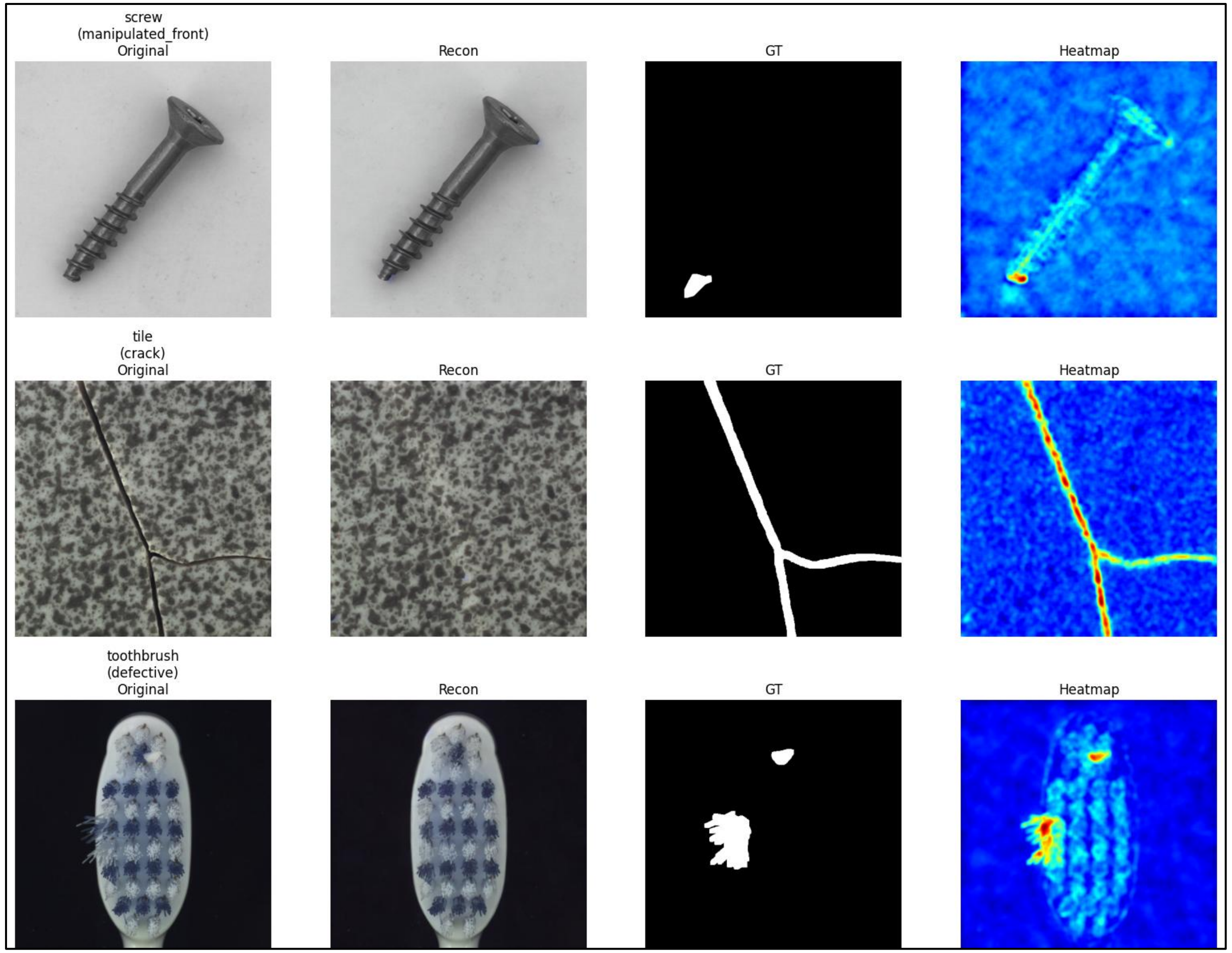
results

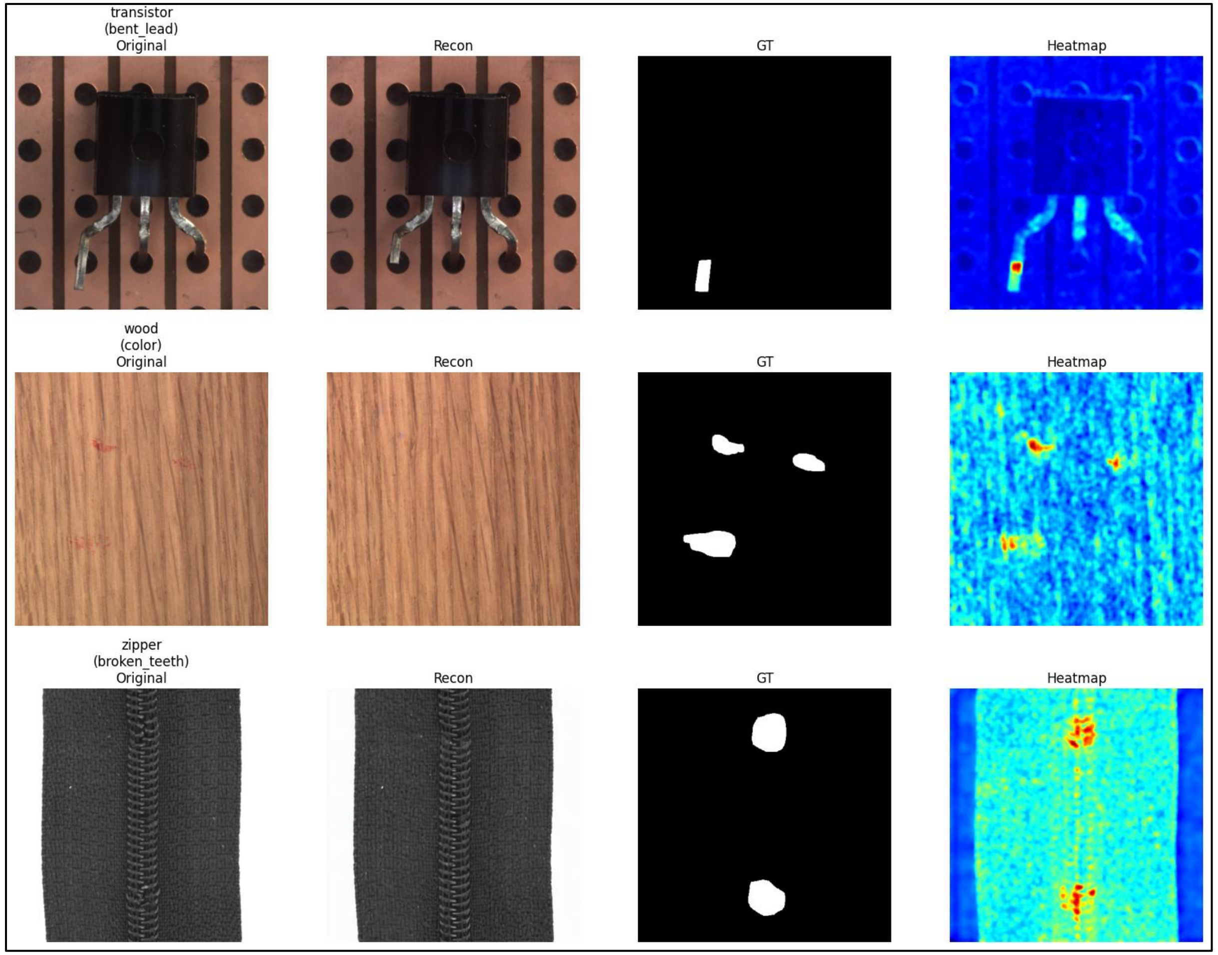




Reconstruction

results





Discussion

Analysis

- Pretrained Diffusion Inpainting 모델만으로도 anomaly 영역이 normal appearance로 효과적으로 복원됨
- 적절한 localization만 주어진다면 학습 없이 reconstruction 기반 anomaly segmentation 가능함

Proposal

- SAM이 생성한 multiple mask candidates에 대해 각각 Diffusion Inpainting 수행
- Reconstruction 결과와 normal reference image 간의 reconstruction score를 계산
- 가장 높은 reconstruction consistency를 가지는 mask를 optimal anomaly mask로 선택

Problem

- Coarse mask를 sam에게 전달해줄 localization model 뭘 써야할지
- musc / winclip / patchcore

Localization model
↓
coarse anomaly region
↓
SAM
↓
mask candidates
↓
Diffusion inpainting
↓
reconstruction score
↓
best mask selection

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